Lick Kast Polarimetry — Exposure Time & S/N Plan

UV-excess QSO candidates (34 targets with g photometry) — imaging & spectropolarimetry

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Methodology (K. Leighly notes 2026-07; E. Glikman's synphot notebook; Chromey,
"To Measure the Sky", sec. 9.5.4)

  1.  Each target is run through the Kast ETC (https://etc.ucolick.org/web_s2n/kast)
      with a flat template and its PS1 g-band AB magnitude + 0.752 mag: the
      polarization optics pass half the flux (m_pol = m + 2.5 log 2).
  2.  ETC settings, identical for all targets:  dichroic d55, slit 2.0 arcsec,
      binning 1x1 pixels, grism 600/4310, grating 600/7500, seeing 1.5",
      airmass 1.1, reference exposure 900 s.
  3.  The ETC's per-pixel object/sky/noise counts vs. wavelength are integrated over the
      PS1 g bandpass via synphot's Observation.effstim() (a throughput-weighted mean),
      then combined with the CCD equation S/N = Nqso / sqrt(Nqso + Nsky + Noise).
  4.  The exposure time reaching the required S/N is the exact solution of the
      CCD equation (Chromey eq. 9.77). Required S/N = 10 ; the 5 floor ("definitely >5") is plotted and tabulate
  5.  Polarimetry penalty: to be sensitive to polarization fraction P, the normal
      exposure time is multiplied by 1/P (rough rule per notes) – x10 for P = 10%,
      x100 for P = 1%. Note P is NOT the scattered fraction: scattering with
      unfavorable geometry can cancel to zero net polarization.
  6.  Spectropolarimetry has no filter to integrate over – every wavelength bin in
      the ETC download needs its own adequate S/N. The blue-grism side only
      (3150-5400 A, before the d55 dichroic split) is evaluated per pixel with the
      same CCD equation; the reported exposure time is the larger of two joint
      requirements: 50% of pixels reach S/N >= 10, and 90% of
      pixels reach S/N >= 5.

Sample summary at S/N = 10, imaging polarimetry (PS1 g)

      P = 10%: 0 targets within 1 h,  0 within 2 h,  0 within 4 h
      P = 1%:  0 targets within 2 h,  0 within 4 h,  0 within 8 h

Sample summary, spectropolarimetry (blue side, joint 50% S/N >= 10 / 90% S/N >= 5)

      P = 10%: 0 targets within 1 h,  0 within 2 h,  0 within 4 h
      P = 1%:  0 targets within 2 h,  0 within 4 h,  0 within 8 h

Caveats:  g magnitudes are observed-frame, not corrected for Galactic
extinction; the 1/P rule is approximate (a 3-sigma detection criterion via
sigma_P = sqrt(2)/(S/N) scales as 1/P^2) – confirm convention before the
proposal; several candidates lie at z < 0.5 (W2M rows) – vet before targeting;
spectropolarimetry uses one joint exposure time (not split by ideal/floor
page) since both S/N criteria are solved together – see the dedicated
spectropolarimetry page below.
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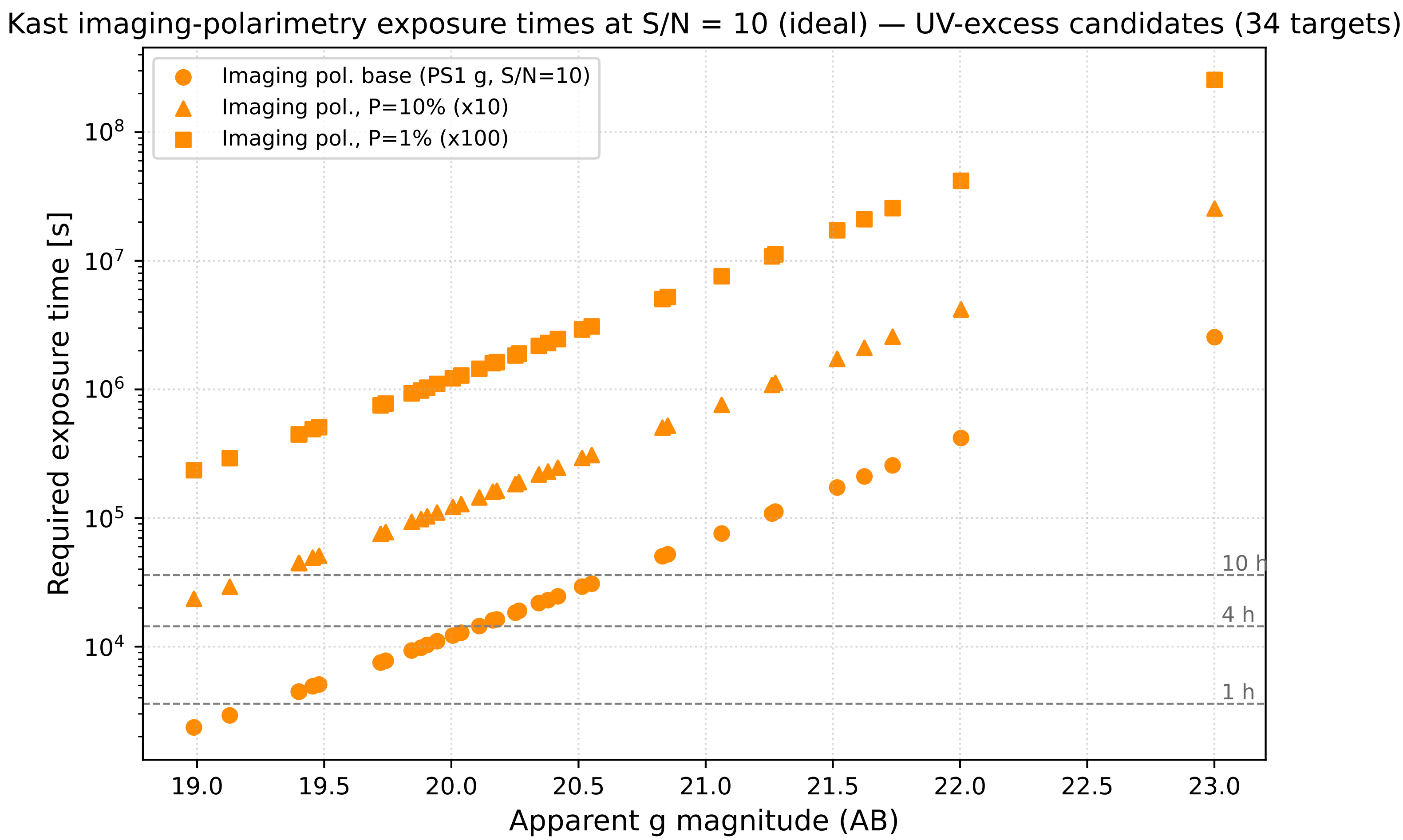
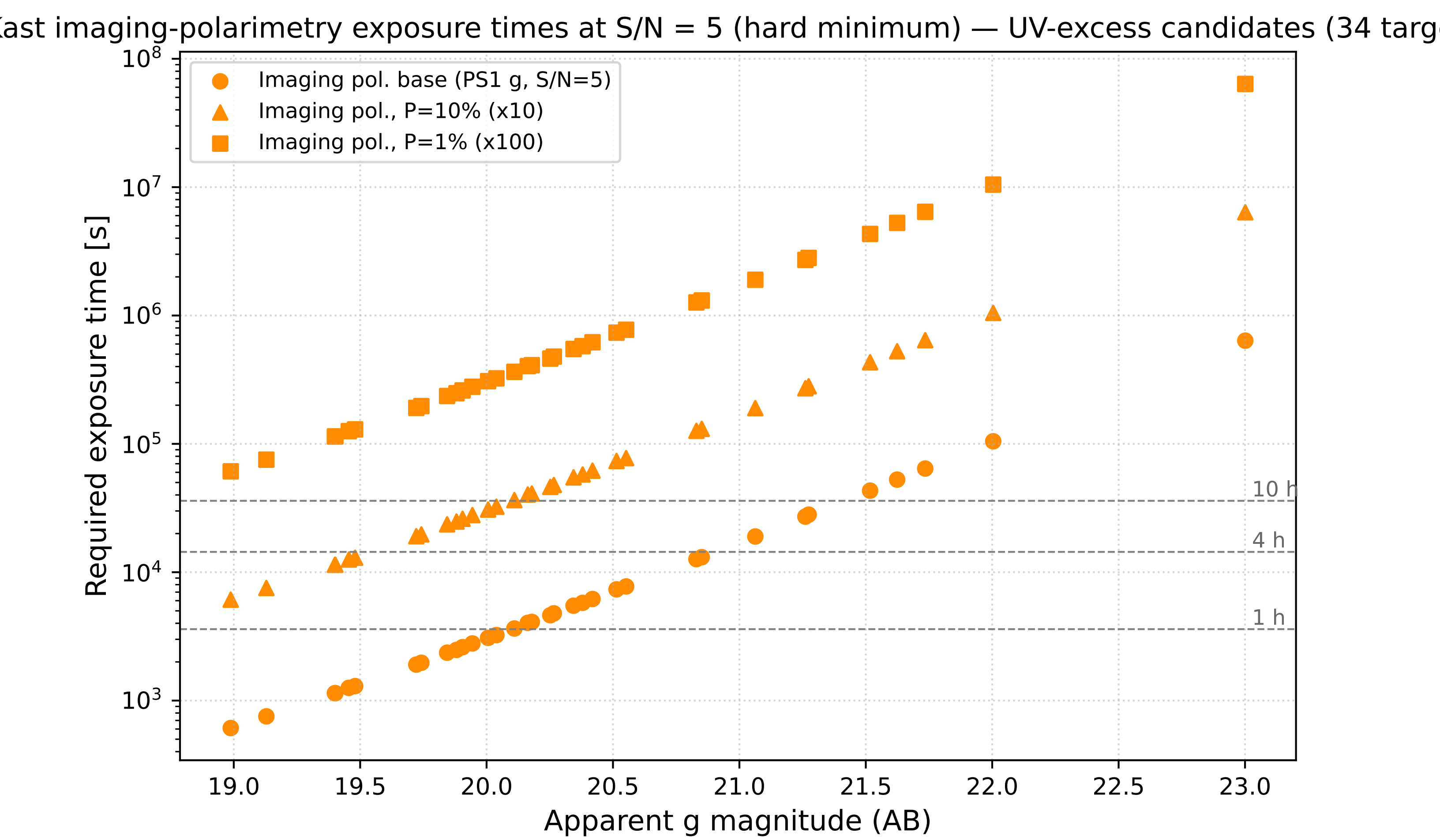

Kast imaging-polarimetry exposure times at S/N = 10 (ideal) — UV-excess candidates (34 targets)
Imaging pol. base (PS1 g, S/N=10)
Imaging pol., P=10% (x10)
Imaging pol., P=1% (x100)
Required exposure time [s]
Apparent g magnitude (AB)
10 h
4 h
1 h
10^8
10^7
10^6
10^5
10^4
19.0
19.5
20.0
20.5
21.0
21.5
22.0
22.5
23.0

Per-target imaging-polarimetry exposure times at S/N = 10 (ideal) (brightest first; PS1 g band)

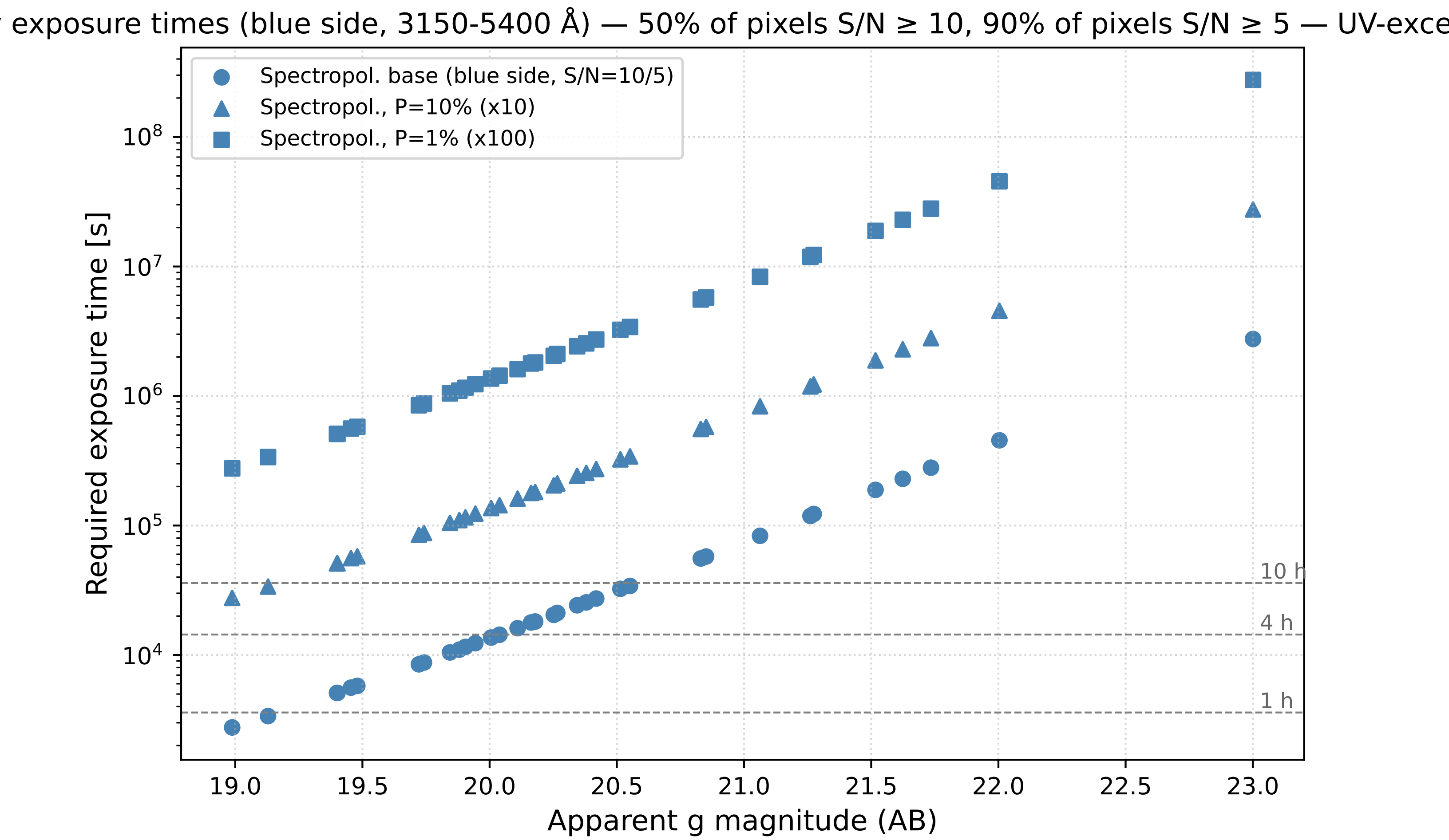
Target	g (AB)	ETC mag	Img base	Img P=10%	Img P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	39.2 m	6.5 h	65.4 h	14
J150746.11+093731.2	19.13	19.88	48.7 m	8.1 h	81.1 h	17
TARGETID_39628512285426154	19.40	20.15	74.5 m	12.4 h	124.1 h	25
TARGETID_39633384095352378	19.40	20.15	74.5 m	12.4 h	124.1 h	25
J150356.69+125914.3	19.45	20.21	82.0 m	13.7 h	136.6 h	28
J144809.53+054019.2	19.48	20.23	84.7 m	14.1 h	141.1 h	29
TARGETID_39628516710418961	19.72	20.47	2.1 h	20.9 h	208.8 h	42
J150357.14+041856.3	19.74	20.49	2.2 h	21.6 h	215.8 h	44
J141354.73+110047.5r	19.84	20.60	2.6 h	25.9 h	259.1 h	52
J152259.51+032347.6r	19.88	20.63	2.7 h	27.2 h	272.4 h	55
J143027.14+152514.8	19.91	20.66	2.9 h	28.6 h	286.4 h	58
J134039.67+051419.7	19.94	20.70	3.1 h	30.6 h	306.3 h	62
J150152.94+020604.9	20.01	20.76	3.4 h	33.9 h	338.9 h	68
J115449.31+112705.8	20.04	20.79	3.6 h	35.7 h	356.5 h	72
TARGETID_39633426973720698	20.11	20.86	4.0 h	40.1 h	401.5 h	81
J110018.80+080455.3	20.16	20.92	4.4 h	44.5 h	444.7 h	89
J132210.17+084232.9	20.18	20.93	4.5 h	45.2 h	452.4 h	91
TARGETID_39633267170738647	20.25	21.00	5.1 h	51.0 h	510.0 h	102
J150013.42+123645.2	20.27	21.02	5.3 h	52.8 h	527.8 h	106
TARGETID_39633136459448983	20.34	21.10	6.1 h	60.6 h	605.7 h	122
TARGETID_39633149587623462	20.38	21.13	6.4 h	63.8 h	638.0 h	128
TARGETID_39633325387681000	20.42	21.17	6.8 h	68.4 h	683.7 h	137
TARGETID_39633396137198490	20.51	21.27	8.1 h	81.3 h	813.4 h	163
TARGETID_39633432401152732	20.55	21.30	8.6 h	85.7 h	857.0 h	172
TARGETID_39633325958105931	20.83	21.58	14.0 h	140.1 h	1400.7 h	281
TARGETID_39633162619324602	20.85	21.60	14.5 h	145.1 h	1451.0 h	291
TARGETID_39633387157195822	21.06	21.81	21.1 h	210.5 h	2105.3 h	422
TARGETID_39632946793025886	21.26	22.01	30.1 h	300.8 h	3007.8 h	602
TARGETID_39633311366122701	21.27	22.03	31.2 h	311.7 h	3117.4 h	624
J140658.28+161142.6	21.52	22.27	48.0 h	479.6 h	4795.9 h	960
TARGETID_39628379145637914	21.62	22.38	58.5 h	584.7 h	5847.1 h	1170
TARGETID_39628346081936841	21.73	22.49	71.3 h	713.2 h	7131.8 h	1427
TARGETID_39633085267968176	22.00	22.76	116.3 h	1163.0 h	11629.7 h	2326
J143404.58+093528.5	23.00	23.75	707.6 h	7076.3 h	70762.8 h	14153



Xast imaging-polarimetry exposure times at S/N = 5 (hard minimum) — UV-excess candidates (34 targe
Imaging pol. base (PS1 g, S/N=5)
Imaging pol., P=10% (x10)
Imaging pol., P=1% (x100)
10 h
4 h
1 h
Required exposure time [s]
Apparent g magnitude (AB)

Per-target imaging-polarimetry exposure times at S/N = 5 (hard minimum) (brightest first; PS1 g band)

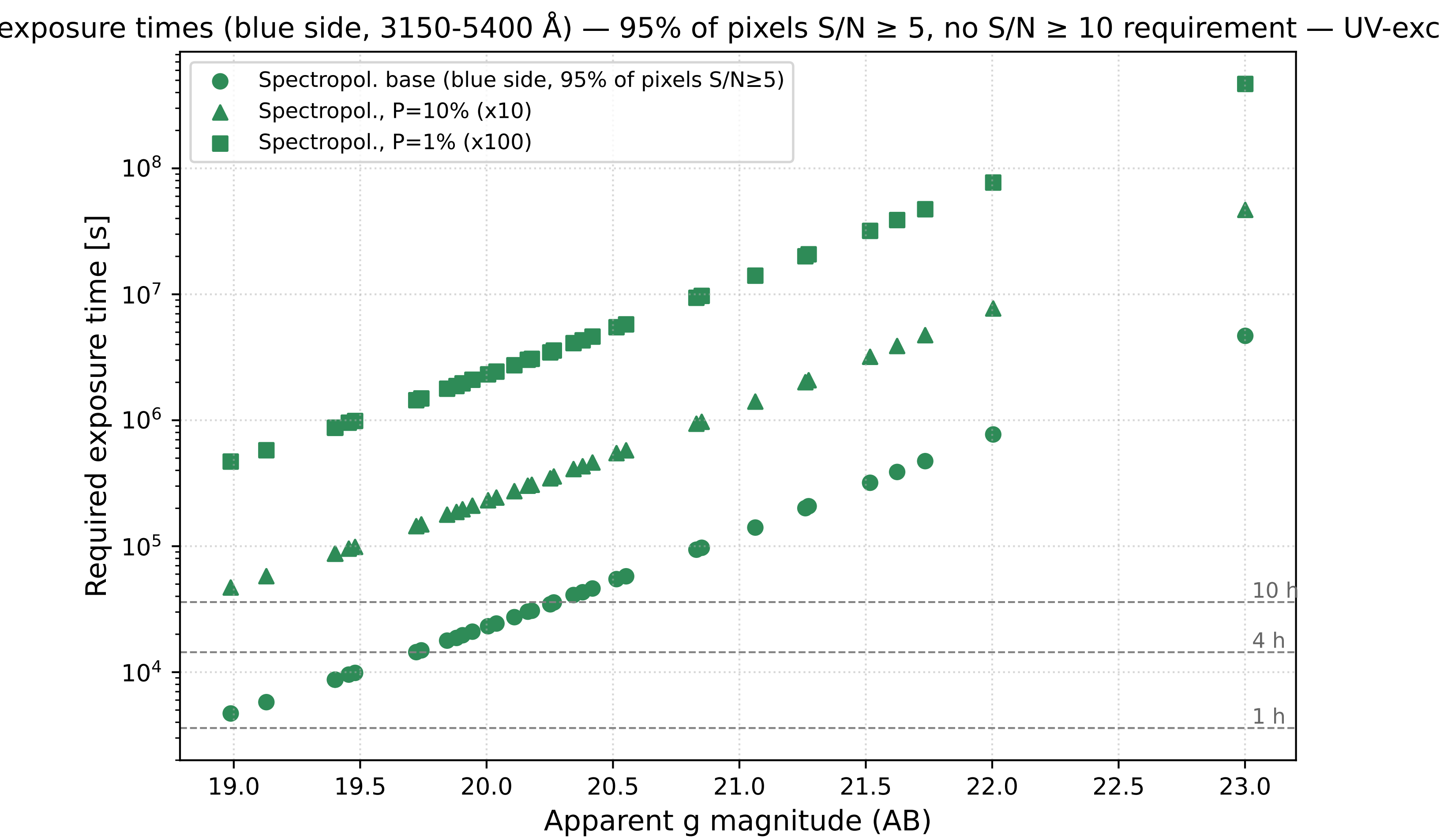
Target	g (AB)	ETC mag	Img base	Img P=10%	Img P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	10.2 m	1.7 h	17.0 h	4
J150746.11+093731.2	19.13	19.88	12.6 m	2.1 h	20.9 h	5
TARGETID_39628512285426154	19.40	20.15	19.0 m	3.2 h	31.7 h	7
TARGETID_39633384095352378	19.40	20.15	19.0 m	3.2 h	31.7 h	7
J150356.69+125914.3	19.45	20.21	20.9 m	3.5 h	34.9 h	7
J144809.53+054019.2	19.48	20.23	21.6 m	3.6 h	36.0 h	8
TARGETID_39628516710418961	19.72	20.47	31.8 m	5.3 h	53.0 h	11
J150357.14+041856.3	19.74	20.49	32.8 m	5.5 h	54.7 h	11
J141354.73+110047.5r	19.84	20.60	39.3 m	6.6 h	65.6 h	14
J152259.51+032347.6r	19.88	20.63	41.3 m	6.9 h	68.9 h	14
J143027.14+152514.8	19.91	20.66	43.4 m	7.2 h	72.4 h	15
J134039.67+051419.7	19.94	20.70	46.4 m	7.7 h	77.4 h	16
J150152.94+020604.9	20.01	20.76	51.3 m	8.6 h	85.5 h	18
J115449.31+112705.8	20.04	20.79	54.0 m	9.0 h	89.9 h	18
TARGETID_39633426973720698	20.11	20.86	60.7 m	10.1 h	101.2 h	21
J110018.80+080455.3	20.16	20.92	67.2 m	11.2 h	112.0 h	23
J132210.17+084232.9	20.18	20.93	68.4 m	11.4 h	113.9 h	23
TARGETID_39633267170738647	20.25	21.00	77.0 m	12.8 h	128.3 h	26
J150013.42+123645.2	20.27	21.02	79.7 m	13.3 h	132.8 h	27
TARGETID_39633136459448983	20.34	21.10	1.5 h	15.2 h	152.3 h	31
TARGETID_39633149587623462	20.38	21.13	1.6 h	16.0 h	160.3 h	33
TARGETID_39633325387681000	20.42	21.17	1.7 h	17.2 h	171.8 h	35
TARGETID_39633396137198490	20.51	21.27	2.0 h	20.4 h	204.2 h	41
TARGETID_39633432401152732	20.55	21.30	2.2 h	21.5 h	215.1 h	44
TARGETID_39633325958105931	20.83	21.58	3.5 h	35.1 h	351.1 h	71
TARGETID_39633162619324602	20.85	21.60	3.6 h	36.4 h	363.7 h	73
TARGETID_39633387157195822	21.06	21.81	5.3 h	52.7 h	527.2 h	106
TARGETID_39632946793025886	21.26	22.01	7.5 h	75.3 h	752.9 h	151
TARGETID_39633311366122701	21.27	22.03	7.8 h	78.0 h	780.3 h	157
J140658.28+161142.6	21.52	22.27	12.0 h	120.0 h	1199.9 h	240
TARGETID_39628379145637914	21.62	22.38	14.6 h	146.3 h	1462.7 h	293
TARGETID_39628346081936841	21.73	22.49	17.8 h	178.4 h	1783.9 h	357
TARGETID_39633085267968176	22.00	22.76	29.1 h	290.8 h	2908.4 h	582
J143404.58+093528.5	23.00	23.75	176.9 h	1769.2 h	17691.7 h	3539



r exposure times (blue side, 3150-5400 Å) — 50% of pixels S/N ≥ 10, 90% of pixels S/N ≥ 5 — UV-exce
Spectropol. base (blue side, S/N=10/5)
Spectropol., P=10% (x10)
Spectropol., P=1% (x100)
Required exposure time [s]
Apparent g magnitude (AB)
10 h
4 h
1 h
10^4
10^5
10^6
10^7
10^8
19.0
19.5
20.0
20.5
21.0
21.5
22.0
22.5
23.0

Per-target spectropolarimetry exposure times (brightest first; blue side only, 3150-5400 Å)
joint requirement: 50% of pixels S/N ≥ target, 90% of pixels S/N ≥ floor

Target	g (AB)	ETC mag	Spec base	Spec P=10%	Spec P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	46.0 m	7.7 h	76.7 h	16
J150746.11+093731.2	19.13	19.88	56.3 m	9.4 h	93.8 h	19
TARGETID_39628512285426154	19.40	20.15	85.0 m	14.2 h	141.7 h	29
TARGETID_39633384095352378	19.40	20.15	85.0 m	14.2 h	141.7 h	29
J150356.69+125914.3	19.45	20.21	1.6 h	15.6 h	155.6 h	32
J144809.53+054019.2	19.48	20.23	1.6 h	16.1 h	160.6 h	33
TARGETID_39628516710418961	19.72	20.47	2.4 h	23.5 h	235.5 h	48
J150357.14+041856.3	19.74	20.49	2.4 h	24.3 h	243.2 h	49
J141354.73+110047.5r	19.84	20.60	2.9 h	29.1 h	291.0 h	59
J152259.51+032347.6r	19.88	20.63	3.1 h	30.6 h	305.7 h	62
J143027.14+152514.8	19.91	20.66	3.2 h	32.1 h	321.2 h	65
J134039.67+051419.7	19.94	20.70	3.4 h	34.3 h	343.1 h	69
J150152.94+020604.9	20.01	20.76	3.8 h	37.9 h	379.1 h	76
J115449.31+112705.8	20.04	20.79	4.0 h	39.8 h	398.5 h	80
TARGETID_39633426973720698	20.11	20.86	4.5 h	44.8 h	448.0 h	90
J110018.80+080455.3	20.16	20.92	5.0 h	49.6 h	495.6 h	100
J132210.17+084232.9	20.18	20.93	5.0 h	50.4 h	504.0 h	101
TARGETID_39633267170738647	20.25	21.00	5.7 h	56.7 h	567.4 h	114
J150013.42+123645.2	20.27	21.02	5.9 h	58.7 h	587.0 h	118
TARGETID_39633136459448983	20.34	21.10	6.7 h	67.3 h	672.8 h	135
TARGETID_39633149587623462	20.38	21.13	7.1 h	70.8 h	708.2 h	142
TARGETID_39633325387681000	20.42	21.17	7.6 h	75.8 h	758.5 h	152
TARGETID_39633396137198490	20.51	21.27	9.0 h	90.1 h	901.0 h	181
TARGETID_39633432401152732	20.55	21.30	9.5 h	94.9 h	949.0 h	190
TARGETID_39633325958105931	20.83	21.58	15.5 h	154.6 h	1546.0 h	310
TARGETID_39633162619324602	20.85	21.60	16.0 h	160.1 h	1600.9 h	321
TARGETID_39633387157195822	21.06	21.81	23.1 h	231.3 h	2313.2 h	463
TARGETID_39632946793025886	21.26	22.01	32.9 h	329.4 h	3294.3 h	659
TARGETID_39633311366122701	21.27	22.03	34.1 h	341.3 h	3413.3 h	683
J140658.28+161142.6	21.52	22.27	52.4 h	523.5 h	5235.3 h	1048
TARGETID_39628379145637914	21.62	22.38	63.8 h	637.5 h	6375.3 h	1276
TARGETID_39628346081936841	21.73	22.49	77.7 h	776.8 h	7767.8 h	1554
TARGETID_39633085267968176	22.00	22.76	126.4 h	1264.0 h	12639.9 h	2528
J143404.58+093528.5	23.00	23.75	765.9 h	7658.9 h	76589.1 h	15318


exposure times (blue side, 3150-5400 Å) — 95% of pixels S/N ≥ 5, no S/N ≥ 10 requirement — UV-exc
Spectropol. base (blue side, 95% of pixels S/N≥5)
Spectropol., P=10% (x10)
Spectropol., P=1% (x100)
Required exposure time [s]
Apparent g magnitude (AB)
10 h
4 h
1 h
10^8
10^7
10^6
10^5
10^4
19.0
19.5
20.0
20.5
21.0
21.5
22.0
22.5
23.0

Per-target spectropolarimetry exposure times (brightest first; blue side only, 3150-5400 Å)
single requirement: 95% of pixels S/N ≥ 5 (no S/N ≥ 10 requirement)

Target	g (AB)	ETC mag	Spec95 base	Spec95 P=10%	Spec95 P=1%	N exp (10%)
J204626.10+002337.6	18.99	19.74	78.3 m	13.0 h	130.5 h	27
J150746.11+093731.2	19.13	19.88	1.6 h	16.1 h	160.5 h	33
TARGETID_39628512285426154	19.40	20.15	2.4 h	24.2 h	242.1 h	49
TARGETID_39633384095352378	19.40	20.15	2.4 h	24.2 h	242.1 h	49
J150356.69+125914.3	19.45	20.21	2.7 h	26.6 h	265.7 h	54
J144809.53+054019.2	19.48	20.23	2.7 h	27.4 h	274.1 h	55
TARGETID_39628516710418961	19.72	20.47	4.0 h	40.1 h	400.7 h	81
J150357.14+041856.3	19.74	20.49	4.1 h	41.4 h	413.8 h	83
J141354.73+110047.5r	19.84	20.60	4.9 h	49.4 h	494.4 h	99
J152259.51+032347.6r	19.88	20.63	5.2 h	51.9 h	519.2 h	104
J143027.14+152514.8	19.91	20.66	5.5 h	54.5 h	545.3 h	110
J134039.67+051419.7	19.94	20.70	5.8 h	58.2 h	582.3 h	117
J150152.94+020604.9	20.01	20.76	6.4 h	64.3 h	642.9 h	129
J115449.31+112705.8	20.04	20.79	6.8 h	67.6 h	675.7 h	136
TARGETID_39633426973720698	20.11	20.86	7.6 h	75.9 h	759.2 h	152
J110018.80+080455.3	20.16	20.92	8.4 h	83.9 h	839.4 h	168
J132210.17+084232.9	20.18	20.93	8.5 h	85.4 h	853.6 h	171
TARGETID_39633267170738647	20.25	21.00	9.6 h	96.0 h	960.4 h	193
J150013.42+123645.2	20.27	21.02	9.9 h	99.3 h	993.4 h	199
TARGETID_39633136459448983	20.34	21.10	11.4 h	113.8 h	1137.7 h	228
TARGETID_39633149587623462	20.38	21.13	12.0 h	119.7 h	1197.4 h	240
TARGETID_39633325387681000	20.42	21.17	12.8 h	128.2 h	1282.0 h	257
TARGETID_39633396137198490	20.51	21.27	15.2 h	152.2 h	1521.8 h	305
TARGETID_39633432401152732	20.55	21.30	16.0 h	160.3 h	1602.5 h	321
TARGETID_39633325958105931	20.83	21.58	26.1 h	260.6 h	2606.4 h	522
TARGETID_39633162619324602	20.85	21.60	27.0 h	269.9 h	2699.2 h	540
TARGETID_39633387157195822	21.06	21.81	39.1 h	390.6 h	3905.6 h	782
TARGETID_39632946793025886	21.26	22.01	55.7 h	556.8 h	5568.1 h	1114
TARGETID_39633311366122701	21.27	22.03	57.7 h	577.0 h	5769.9 h	1154
J140658.28+161142.6	21.52	22.27	88.6 h	885.9 h	8859.0 h	1772
TARGETID_39628379145637914	21.62	22.38	107.9 h	1079.3 h	10792.8 h	2159
TARGETID_39628346081936841	21.73	22.49	131.6 h	1315.5 h	13155.2 h	2632
TARGETID_39633085267968176	22.00	22.76	214.2 h	2142.3 h	21423.1 h	4285
J143404.58+093528.5	23.00	23.75	1300.0 h	12999.9 h	129999.3 h	26000